

ON A FIRST-ORDER HYPERBOLIC APPROXIMATION OF THE CAHN-HILLIARD EQUATION.

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Diffusion equations in physics and mechanics

A large part of continuum physics is written in terms of parabolic operators:

$$\frac{\partial \theta}{\partial t} = \operatorname{div} (K(x, \theta) \nabla \theta) \quad (\text{Fourier's law of heat conduction})$$

$$\begin{cases} \frac{\partial \rho}{\partial t} + \operatorname{div} (\rho \mathbf{u}) = 0 \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = \operatorname{div} (\mathbb{S}(\mathbf{u}, \nabla \mathbf{u})) \end{cases} \quad (\text{Navier–Stokes' system})$$

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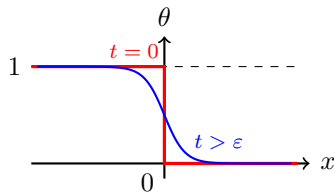
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- Require **postulating a constitutive law**: $\mathbf{q} = -K \nabla \theta$, $\mathbf{j} = -D \nabla c$, etc...
- The flux responds **instantaneously** to the gradient.

Instantaneous propagation – Paradox of the heat equation

$$\frac{\partial \theta}{\partial t} = K \Delta \theta, \quad \theta(x, 0) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\implies \theta(x, t) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{Kt}}\right).$$

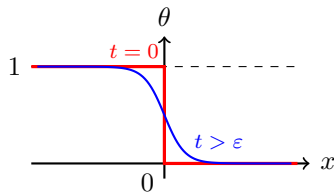


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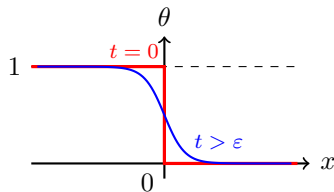
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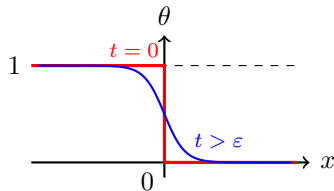
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- **Incompatibility** with relativity (speed of light is the limit)
- Wrong physics at very short times / extreme conditions.

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[1] L. Tisza, "Transport phenomena in Helium II," *Nature*, vol. 141, p. 913, 1938.

[2] L. Landau, "On the Theory of Superfluidity of Helium II," *Journal of Physics USSR*, vol. 11, no. 1, pp. 91–92, 1947.

[3] Peshkov, V. P. "Second sound in helium II." *Sov. Phys. JETP* 11.3 (1960): 580-584.

The Cattaneo model of relativistic heat conduction

1) Take the classic linear heat equation

$$\frac{\partial \theta}{\partial t} - K \Delta \theta = 0 \quad \Longleftrightarrow \quad \begin{cases} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} = 0, \\ \mathbf{q} = -K \nabla \theta. \end{cases}$$

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2) Replace the **instantaneous** constitutive law by a *relaxed* version

$$\mathbf{q} = -K \nabla \theta \quad \Longrightarrow \quad \tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -K \nabla \theta, \quad 0 < \tau \ll 1.$$

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We obtain the system

$$\begin{aligned} \frac{\partial \theta}{\partial t} + \nabla \cdot \mathbf{q} &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \frac{K}{\tau} \nabla \theta &= -\frac{1}{\tau} \mathbf{q}. \end{aligned}$$

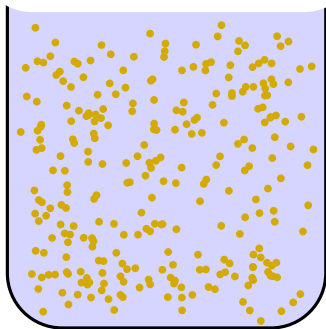
Cahn-Hilliard equations (1958)

The Cahn-Hilliard equation writes

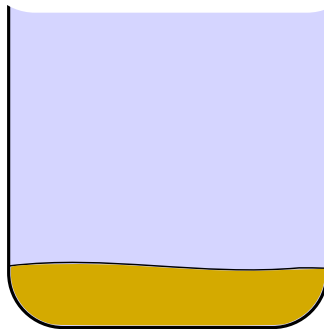
$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc.

Phase separation: example of sedimentation



Dispersed phase



Separated phases

Conservative form and chemical potential

The Cahn-Hilliard equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

is derived from the energy functional

$$F[c, \nabla c] = \int_{\Omega} f(c, \nabla c) d\Omega, \quad \text{where} \quad f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 .$$

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through the **postulate** of a generalized Fick's law

$$\frac{\partial c}{\partial t} = \operatorname{div} (\nabla \mu), \quad \mu = \frac{\partial f}{\partial c} - \operatorname{div} \left(\frac{\partial f}{\partial \nabla c} \right)$$

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Lyapunov functional:
$$\frac{\partial F}{\partial t} = - \int_{\Omega} \|\nabla \mu\|^2 d\Omega \leq 0 + B.C$$

Modified action functional

Based on the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \left(\frac{c^2 - 1}{4} \right)^2 + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2 - \left(\frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla \varphi\|^2 + \frac{\alpha}{2} (c - \varphi)^2 \right)$$

- φ is a new variable substituting c as the order parameter.
- $\alpha \gg 1$ so that $(c - \varphi)$ vanishes in the limit $\alpha \rightarrow +\infty$.
- $\beta \ll 1$ is a small parameter.

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Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

$\Rightarrow$ 2nd-order PDE, no 4th-order terms

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

Euler-Lagrange equation for φ

$$\mathcal{L}\left(\mathbf{c}, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (\mathbf{c} - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

For φ , we simply write the Euler-Lagrange equations.

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \varphi_t} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla \varphi} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}.$$

which gives

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) = \alpha (c - \varphi) \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far, instead of

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c),$$

we have obtained the following system of two 2nd order PDEs

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- Equation (I) is reminiscent of heat equation.
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- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.
- Equation (II) is a hyperbolic wave equation with right-hand side. $\Rightarrow$ Order reduction.

Cattaneo relaxation for equation (I)

$$\begin{aligned}\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \nabla \mu &= -\frac{1}{\tau} \mathbf{q},\end{aligned}$$

- $\tau \ll 1$ is a relaxation time.
- c is still a conserved quantity in this framework.

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

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Therefore (II) becomes

$$\begin{aligned} \frac{\partial w}{\partial t} - \operatorname{div}(\gamma \mathbf{p}) &= -\alpha(\varphi - c), \\ \frac{\partial \varphi}{\partial t} &= \frac{1}{\beta} w, \\ \frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w &= 0. \end{aligned}$$

Final system approximating the Cahn-Hilliard equations

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{1}{\beta} w \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

Hyperbolicity

System admits a full set of real eigenvalues (with $\alpha > 1$) given by

$$\lambda_{1,2} = \pm \frac{\sqrt{3c^2 - 1 + \alpha}}{\sqrt{\tau}}, \quad \lambda_{3,4} = \pm \frac{\sqrt{\gamma}}{\sqrt{\beta}}, \quad \lambda_{5-9} = 0$$

and a corresponding full set of linearly independent eigenvectors.

$\Rightarrow$ System is strongly hyperbolic.

Lyapunov Functional

Proposition

The proposed hyperbolic Cahn-Hilliard system admits the following Lyapunov functional

$$E = \int_{\mathcal{D}} e(c, \varphi, \mathbf{q}, \mathbf{p}, w) \, d\Omega,$$
$$e(c, \varphi, \mathbf{q}, \mathbf{p}, w) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi}$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w}$$

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$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

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$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

$$\Rightarrow \frac{\partial e}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial c} \frac{\partial e}{\partial \mathbf{q}} - \frac{\partial e}{\partial \mathbf{p}} \frac{\partial e}{\partial w} \right) = - \left\| \frac{\partial e}{\partial \mathbf{q}} \right\|^2 \leq 0,$$

Numerical methods

In order to solve the model numerically and also compare it with reference solutions, we propose here:

- ① A numerical scheme for the original Cahn-Hilliard equation based on 4th order semi-implicit conservative finite differences
- ② Explicit MUSCL-Hancock for the hyperbolic approximation.

Implicit conservative finite differences for CH

We propose here a semi-implicit conservative in order to solve numerically the original Cahn-Hilliard equations. We rewrite the latter as follows

$$\frac{\partial c}{\partial t} - \operatorname{div}(\mathbf{F}) + \gamma \Delta^2 c = 0$$

where $\mathbf{F}$ is the flux given by

$$\mathbf{F} = \chi(c) \nabla c, \quad \chi(c) = 3c^2 - 1$$

The scheme writes

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta \Delta_h c_{i,j}^{n+1}.$$

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using conservative finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^n (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^n \simeq \frac{1}{12} (7\chi_{i,j}^n - \chi_{i-1,j}^n + 7\chi_{i+1,j}^n - \chi_{i+2,j}^n) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12\Delta x} (15c_{i,j}^{n+1} - 15c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} - c_{i-1,j}^{n+1}) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

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(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

These are 4th order approximations.

Discretization of the bi-Laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{1}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{1}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

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$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{1}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{1}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta\Delta_h c_{i,j}^{n+1}.$$

is then solved using GMRES (Matrix is not symmetric ...)

Numerical method for hyperbolic approximation

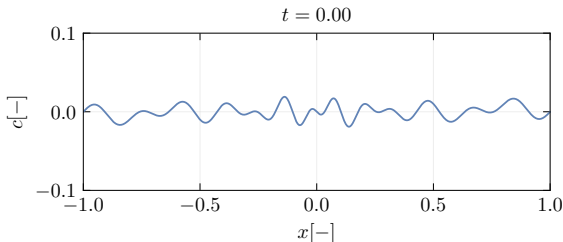
- Explicit second-order MUSCL-Hancock scheme
- We use either FORCE or Rusanov approximate Riemann solvers (One could also implement a Roe solver)

Spinodal decomposition

We suggest the following initial data

$$c(x) = \begin{cases} 0.01 \left((\sin(10\pi(1+x))) - \sin(10\pi(1+x)^2) \right), & \text{if } x \in [-1, 0] \\ -0.01 \left((\sin(10\pi(1-x))) - \sin(10\pi(1-x)^2) \right), & \text{if } x \in [0, 1]. \end{cases}$$

This function is built in such a way that it is C^∞ over $[-1, 1]$ as well as over $\mathbb{R}$ by periodic prolongation.



Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

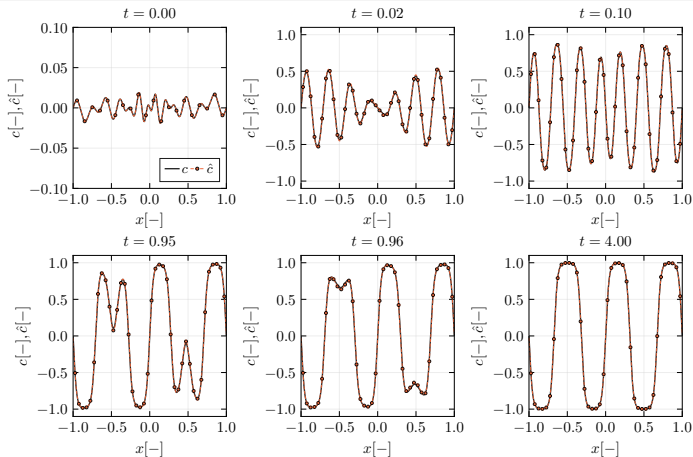


Figure 1: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Spinodal decomposition - Lyapunov function evolution ($\simeq 6b$ iterations)

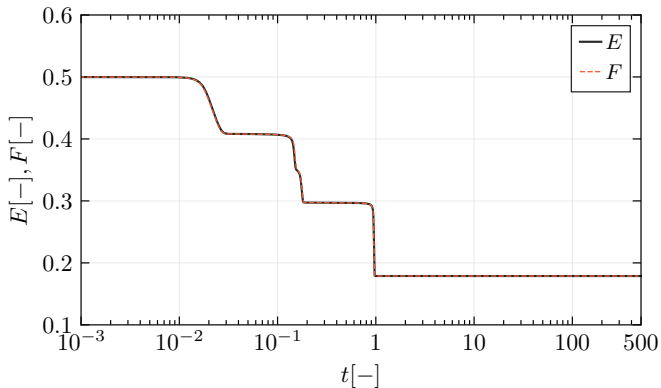


Figure 2: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Lyapunov function evolution ($\simeq 30b$ iterations)

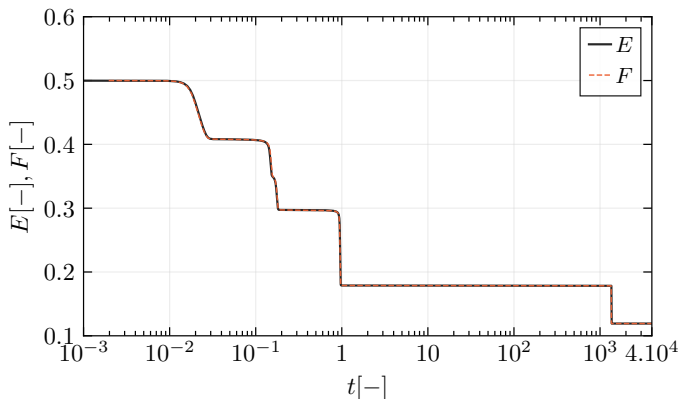


Figure 3: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

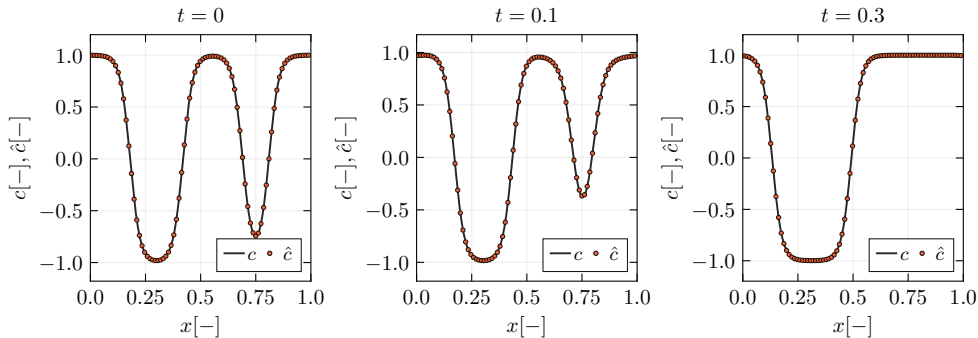
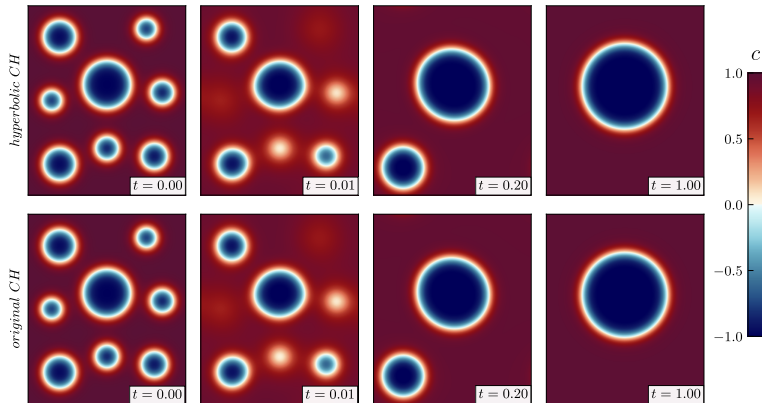


Figure 4: Comparison of the numerical solutions for hyperbolic Cahn-Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)



Ostwald Ripening in 2D: horizontal Cuts

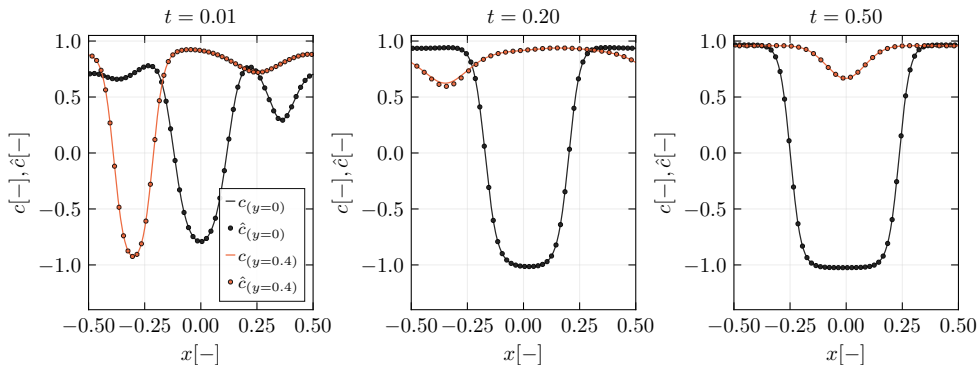


Figure 5: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Honorable mentions

- Analysis of the model & convergence proof was recently done by Gieselmann & Ranocha (2026).
- In particular they provide a scaling asymptotic scaling of parameters given.

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Conclusion and Perspectives

Conclusions

- Diffusion equations can be approximated by hyperbolic models.
- Not every hyperbolic approximation conserves properties.
- In both cases, the approach was validated theoretically and numerically using standard numerical methods.

Perspectives

- Extension to Navier-Stokes-Cahn-Hilliard models, potentially in stiff regimes (Work in progress with L.del Rio (inria Bdx) and A.Lucca (U.Trento))
- Semi-implicit discretization to circumvent stiffness (Work in progress with Andreu Martorell and Pep Mulet (Univ Valencia)).
- Adaptive time-stepping strategies for varying time scale solutions.

Thank you for your attention !

[2] Dhaouadi, Firas, Dumbser, Michael, & Gavrilyuk, S. A first-order hyperbolic reformulation of the Cahn–Hilliard equation. " Proceedings of the Royal Society A (Vol. 481, No. 2312, p. 20240606).

And references therein.

All the codes are published alongside the paper.